Universidade de Brasília
Faculdade de Tecnologia
Departamento de Engenharia Mecânica

Gated Data-Space Inversion for Reservoir Prediction

Thiago Tomás de Paula

DISSERTAÇÃO DE MESTRADO
PROGRAMA DE PÓS-GRADUAÇÃO EM CIÊNCIAS MECÂNICAS

Brasília
2026

Universidade de Brasília
Faculdade de Tecnologia
Departamento de Engenharia Mecânica

Agregação de Modelos Adaptativa Aplicada a Otimização de Reservatórios

Thiago Tomás de Paula

Dissertação de Mestrado submetida ao Programa de Pós-Graduação em Ciências Mecânicas como requisito parcial para obtenção do grau de Mestre.

Orientador: Prof. Dr. Eugênio Libório Feitosa Fortaleza

Brasília
2026

FICHA CATALOGRÁFICA

P324g

Paula, Thiago Tomás de.

Gated Data-Space Inversion for Reservoir Prediction / Thiago Tomás de Paula; orientador Eugênio Libório Feitosa Fortaleza. -- Brasília, 2026.

18 p.

Dissertação de Mestrado (Programa de Pós-Graduação em Ciências Mecânicas) -- Universidade de Brasília, 2026.

1. . 2. . 3. . I. Fortaleza, Eugênio Libório Feitosa, orient. II. Título.

Universidade de Brasília
Faculdade de Tecnologia
Departamento de Engenharia Mecânica

Gated Data-Space Inversion for Reservoir Prediction

Thiago Tomás de Paula

Dissertação de Mestrado submetida ao Programa de Pós-Graduação em Ciências Mecânicas como requisito parcial para obtenção do grau de Mestre.

Trabalho aprovado. Brasília, 15 de julho de 2026:

Prof. Dr. Eugênio Libório Feitosa Fortaleza
UnB/FT/ENM
Orientador

Examinador interno

Examinador interno

Abstract

Guiding questions

- What is the reason for writing the thesis?
- What are the current approaches and gaps in the literature?
- What are your research question(s) and aims?
- Which methodology have you used?
- What are the main findings?
- What are the main conclusions and implications?

Keywords: ; ; .

Resumo

Perguntas orientadoras

- Qual é a motivação para escrever a dissertação?
- Quais são as abordagens atuais e lacunas na literatura?
- Quais são as suas perguntas de pesquisa e objetivos?
- Qual metodologia você utilizou?
- Quais são os principais resultados?
- Quais são as principais conclusões e implicações?

Palavras-chave: ; ; .

Contents

1 Introduction

Guiding questions

- What is your original contribution? Why should the examiner care about your research?
- What is the thesis problem statement?
- What do you (not) hope to achieve?
- What are the research questions and hypotheses?
- What are your epistemological and ontological positions?
- What is your contribution to the field?
- How is the thesis laid out?

PhD introduction at a glance: A typical PhD thesis introduction follows the following

Element	**Purpose**	**Position**	**Typical Length**
Background Context	Set the scene for your research	Opening section	500-1,000 words
Research Problem	Define the specific gap or issue	After context	300-500 words
Research Questions	State your precise questions	After problem	200-400 words
Significance	Explain why this research matters	After questions	300-500 words
Methodology Overview	Brief summary of approach	After significance	200-400 words
Thesis Outline	Chapter-by-chapter roadmap	Final section	300-500 words

format:

- Introduction to the introduction: a short version (of only a few paragraphs) of the thesis' aims, research questions, contribution, objectives and findings.
- State the overarching topic and aims of the thesis in more detail
- Provide a brief review of the literature related to the topic (this will be very brief if you have a separate literature review chapter)
- Define the terms and scope of the topic
- Critically evaluate the current state of the literature on that topic and identify your gap
- Outline why the research is important and the contribution that it makes
- Outline your epistemological and ontological position
- Clearly outline the research questions and problem(s) you seek to address
- State the hypotheses (if you are using any)
- Detail the most important concepts and variables
- Briefly describe your methodology
- Discuss the main findings
- Discuss the layout of the thesis

Once you have finished your thesis, ask yourself the following questions:

- Does the first line of the introduction discuss the problem that your thesis is addressing and the contribution that it is making?
- Does the introduction provide an overview of the thesis and end with a brief discussion on the content of each chapter?
- Does the introduction make a case for the research?
- Have the research questions/problems/hypotheses been clearly outlined (preferably early on)?

1.1 Work overview

1.2 Background context

1.3 Research problem

1.4 Research questions

1.5 Significance

1.6 Methodology overview

1.7 Dissertation outline

2 Literature Review

2.1 Chapter outline

Aims and structure

According to Hart (2018), the job of a literature review is to show five things (if you're using our free PhD Writing Template, you may recognise these):

- What has been written on your topic
- Who the key authors are and what the key works are
- The main theories and hypotheses
- The main themes that exist in the literature
- Gaps and weaknesses that your study will then help fill

A literature review has three objectives:

- Summarise what has already been discussed in your field, both to demonstrate that you understand your field and to show how your study relates to it.
- Highlight gaps, problems or shortcomings in existing research to show the original contribution that your thesis makes.
- Identify important studies, theories, methods or theoretical frameworks that can be applied in your research.

There are nine steps involved in conducting a literature review:

- Pick a broad topic
- Find the way in
- Who's saying what and when
- Take notes
- Narrow down the field
- Narrow down the sources
- Snowball
- Think about questions that haven't been asked
- Write early, write quickly and write relevantly

2.2 Chapter summary

Key takeaways

3 Theory framework

3.1 Chapter outline

Aims and structure

The theory framework is the scaffolding upon which your thesis is built. When you're done writing your theory framework chapter or section, your reader should be able to answer these questions:

- What theoretical concepts are used in the research? What hypotheses, if any, are you using?
- Why have you chosen this theory?
- What are the implications of using this theory?
- How does the theory relate to the existing literature, your problem statement and your epistemological and ontological positions?
- How has this theory has been applied by others in similar contexts? What can you learn from them and how do you differ?
- How do you apply the theory and measure the concepts (with reference to the literature review/problem statement)?
- What is the relationship between the various elements and concepts within the model? Can you depict this visually?

That means that a theory framework can take different forms:

- It can state the theoretical assumptions underpinning the study.
- It can connect the empirical data to existing knowledge.
- It can allow you to come up with propositions, concepts or hypotheses that you can use to answer 'how' and 'why' questions.

3.2 Elements of reservoir engineering

3.2.1 Reservoir formation

3.2.2 Reservoir simulation

3.3 Kalman Filtering

3.4 Data-Space Inversion

3.5 Chapter summary

Key takeaways

4 Methods

The job of a methods chapter is:

- To summarise, explain and recount how you answered your research questions and to explain how this relates to the methods used by other scholars in similar contexts and similar studies
- To discuss - in detail - the techniques you used to collect the data used to answer your research questions
- To discuss why the techniques are relevant to the study's aims and objectives
- To explain how you used them

Your reader should be able to answer the following questions when they're done reading it:

- What did you do to achieve the research aims?
- Why did you choose this particular approach over others?
- How does it relate to your epistemological and ontological positions?
- What tools did you use to collect data and why? What are the implications?
- When did you collect data, and from whom?
- What tools have you used to analyse the data and why? What are the implications? Are there ethical considerations to take into account?

4.1 Chapter outline

Aims and structure

4.2 Reservoir case studies

4.2.1 Egg ensemble

4.2.2 Olympus ensemble

4.2.3 Real field

4.3 Chapter summary

Key takeaways

5 Results

When your reader is finished with this section, they should be able to answer the following questions:

- What are the results of your investigations?
- How do the findings relate to previous studies?
- Was there anything surprising or that didn't work out as planned?
- Are there any themes or categories that emerge from the data?
- Have you explained to the reader why you have reached particular conclusions?
- Have you explained the results?

5.1 Chapter outline

Aims and structure

5.2 Egg ensemble

5.3 Olympus ensemble

5.4 Real field

5.5 Chapter summary

Key takeaways

6 Discussion

6.1 Chapter outline

Aims and structure

You are providing sufficient detail that others can draw their own inferences and construct their own explanations. Think of it as presenting the case for a jury. That means that an empirical discussion should:

- Tell the reader how the data was collected, with reference to the methods chapter/section
- Tell them how they can access it if they wanted to replicate your study
- Discuss what the results look like (using visual aids, such as tables, diagrams, graphs and so on)
- Provide rich summaries of the findings
- Discuss the gaps in the findings and analysis
- Analyse the results
- Discuss the implications of your findings
- Discuss the limitations of the findings

When it comes to writing your discussion chapter, you can start by writing a few sentences that summarise the most important results. One danger when writing discussion sections is that they can be too wordy, offer too much interpretation and lack a clear structure. To avoid this, you should make sure that every element of your discussion section addresses one of the following questions:

- What are the relationships between observations? (The mud-map you developed earlier will help here.)
- Are there any trends and generalisations amongst the results? Are there any exceptions to these?
- What are the causes of, or mechanisms behind, the underlying patterns you have uncovered?
- Do your results agree or disagree with previous work?
- How do your findings relate to the theoretical framework you developed, if applicable?
- How do the findings relate to the hypotheses you developed, if applicable?
- What other explanations could there be for your results? This issue is more pertinent if you are engaging in theory creation/inductive reasoning.
- What do we now know as a result of your research that we didn't know before?

- What is the significance of these findings?
- Why should we care about the findings?

When your discussion chapter is finished, your reader should be able to answer the following questions:

- How do the findings relate to the theory and methods discussed previously?
- Why you have reached particular conclusions?
- How do your findings relate to the gaps in the literature you identified earlier?
- What implications do the findings have for the discipline and for existing understanding?
- How do the findings relate to your research questions/aims and objectives?

6.2 Summary of Key Findings

- Brief Restatement of Research Findings
- Comparison with Initial Hypotheses or Research Questions

6.3 Interpretation of Findings

- Contextualisation of Results
- Significance and Implications of the Findings

6.4 Evaluation of Existing Theories and Models

- How Your Findings Support or Challenge Previous Work
- Conceptual Contributions of Your Study

6.5 Limitations and Future Research

- Acknowledgment of Study Limitations
- Suggestions for Future Research

6.6 Chapter summary

Key takeaways

7 Conclusion

The job of the conclusion is to:

- Fully and clearly articulate the answer to your research questions
- Discuss how the research is related to your aims and objectives
- Explain the significance of the work
- Outline its shortcomings
- Suggest avenues for future research

It is not the place to introduce new ideas and concepts, or to present new findings. Your job is to reflect back on your original aims and intentions and discuss them in terms of your findings and new expertise.

By the time the reader has finished reading the conclusion, they should be able to answer the following questions:

- Have you briefly recapped the research questions and objectives?
- Have you provided a brief recount of the answer to those questions?
- Have you clearly discussed the significance and implications of those findings?
- Have you discussed the contribution that the study has made?
- Do the claims you are making align with the content of the results and discussion chapters?

References

SCHAUM, D. **Schaum's outline of theory and problems**. 5th. ed. New York: Schaum Publishing, 1956. 204 p. Cit. on p. 18.

(Schaum, 1956)